Olist E-Commerce Analytics Project

Customer Satisfaction, Delivery Performance & Logistics Bottlenecks

Python/Pandas • SQL • Power BI

Business Question

Management has noticed that customer satisfaction has been inconsistent, and they want to understand which factor may be contributing to poor customer experiences.

Overview

This project analyzes Brazilian E-commerce public dataset by olist to identify what factors are contributing to poor customer satisfaction, where delivery problems occur, and what the main bottleneck is. Review score was used as a proxy for customer satisfaction throughout the analysis.

The analysis was conducted using Python/Pandas for data cleaning and feature engineering, SQL for data analysis and hypothesis testing, and Power BI for interactive dashboard development and communication of findings to non-technical users.

Dataset

- Source : [Brazilian E-Commerce Public Dataset by Olist]
- 99441- Orders, 32951 - Products, 112650 - Orders_items, 99224 - Reviews and 99441 - customers
- Tables for this analysis:
    1. Products,
    2. Orders,
    3. Order_items,
    4. Review,
    5. Customers

Tools Used

- Python (pandas, numpy): Data Cleaning and Creating calculated columns
- SQL : Data analysis and Queries
- Power BI : Final Interactive Design

Methodology

- **Python(Pandas, numpy)**

    1. cleaned the datasets
    2. handled missing values
    3. created calculated columns
    4. created Quartile-based categories were created using the 25th percentile (Q1) and 75th percentile (Q3) for price and freight

    During data validation, 13,059 orders were identified where the carrier delivery date was later than the approval date. Rather than removing these records, they were retained and flagged using a Boolean indicator(couriered_before_approval) for further analysis.

- **SQL**

    Six hypotheses are created and tested.

    Hypothesis 1 : Late Delivery may leads lower customer satisfaction

    I compared the Average Review Score between Late and On-time delivery status. I noticed that On-time delivery has a higher avg score (4.29) than Late delivery (2.57).

    The result supports my hypothesis that Late delivery may contribute to lower customer satisfaction.

    Hypothesis 2 : Higher freight cost may increase the lower customer satisfaction

I compared the average review score across all freight categories. I noticed that Average review score decreased from 4.11 for low freight cost to 4.04 for medium freight cost and 3.94 for high freight cost

This result support my hypothesis that higher freight cost may contribute to the lower customer satisfaction

Hypothesis 3 : Higher product price may contribute to lower customer satisfaction

I compared the average review score across all price categories. I noticed that the average review score is relatively the same across all the price categories (4.03). This result does not support my hypothesis which means Higher product price might not contribute to lower customer satisfaction

Hypothesis 4 : longer seller handling time may contribute to lower customer satisfaction?

The seller taking too much time to deliver got lowest avg review score (3.69) while sellers who takes less time to deliver got highest avg score (4.26)

This result supports my hypothesis which means, longer seller handling time may contribute to lower customer satisfaction.

Hypothesis 5 : Extreme delivery delays are mainly caused by the seller or the carrier.

I analyzed the top 1% of orders with the longest delivery times.I noticed that the average carrier shipping time was 53.31 days, while seller handling took only 8.58 days.

This suggests that the carrier/shipping stage is the main bottleneck in extreme delivery delays, accounting for approximately 86% of the total average processing time.

Hypothesis 6 : Which geographic area has the highest average delivery time?

I compared average delivery time across different states.I noticed that RR state has the highest average delivery time of 29.34 days.

This suggests that customers in RR state area may experience longer delivery times, possibly due to distance or logistics constraints

Appendix — Key Results

Analysis	Metric	Result
Delivery	On-time average review	4.29
Delivery	Late average review	2.57
Freight	Low / Medium / High review	4.11 / 4.04 / 3.94
Price	Average review	≈ 4.03
Handling	Shortest / Longest review	4.26 / 3.69
Extreme delays	Orders in top 1%	965
Extreme delays	Avg seller handling	8.58 days
Extreme delays	Avg carrier shipping	53.31 days
Extreme delays	Seller / Carrier contribution	13.87% / 86.13%
Geography	Avg Days by State	RR — 29.34 days

- **Power Bi**
  1. Power Query Editor
  2. relationships between tables
  3. calculated measures
  4. KPI cards
  5. charts
  6. geographic analysis
  7. extreme-delay bottleneck analysis

Relationships between the relevant tables were established using Power Query and the Power BI data model.

Business Recommendation through my hypothesis

1. Focus more on delivery time and reduce late deliveries.

   Late delivery has a lower average review score, so the company should focus on improving delivery time and reducing late deliveries.

2. Optimize shipping costs.

   Higher freight cost is associated with lower average review scores, so the company should look for ways to optimize shipping costs.

3. Reduce seller processing time.

   Orders with longer seller handling time have lower average review scores, so sellers should try to process and hand over orders to the carrier faster.

4. Improve carrier performance

   Carrier shipping takes much longer than seller handling in extreme-delay orders, so the company should focus on reducing the time orders spend in the shipping stage.

5. Investigate high-delay geographic areas

   The company should focus on areas with longer delivery times and find ways to reduce delivery delays.

Conclusion

This analysis investigated several factors that may contribute to lower customer satisfaction in the Olist e-commerce marketplace.

The results show that delivery-related factors were more strongly associated with review scores than product price. Late deliveries had lower average review scores than on-time deliveries, while longer seller handling time was also associated with lower review scores.

The extreme-delay analysis provided an additional operational insight. Among the top 1% longest-delivery orders, carrier shipping accounted for 86.13% of the combined average seller and carrier processing time, compared with 13.87% for seller handling. This identifies the carrier/shipping stage as the main bottleneck in extreme delivery cases.

The geographic analysis also showed differences in delivery performance, with RR state recording the highest average delivery time at 29.34 days in the analyzed result.

One hypothesis : the relationship between product price and customer satisfaction — was not supported, as average review scores remained approximately 4.03 across the price categories.

Overall, the project demonstrates how Python, SQL, and Power BI can be combined to move from raw e-commerce data to actionable business insights.

Dashboard